Early-Phase Clinical Pharmacology — Biostatistics Deliverables Pipeline

Protocol review → CSR draft · every document, dataset, and statistical output · with the hybrid-AI routing overlay

> ***Scope & how to read this.*** *This is the complete biometrics deliverables map for a Phase 1 clinical-pharmacology study (e.g., First-in-Human SAD/MAD with PK/PD, NCA + popPK, safety). 136 deliverables across five stages, each with its owner, inputs, outputs, governing standard, QC method, and how AI is (or is not) used. The at-a-glance tables are below; the* ***full register with every field is the companion `.xlsx`****, and the* ***one-page visual is the `.pdf` map****. Standards are current as of mid-2026 (CDISC SDTMIG v3.4 / ADaMIG v1.3 / Define-XML v2.1; ICH E3, E6(R3), E9/E9(R1), M11; 21 CFR Part 11; ISPE GAMP 5).*

How the pipeline flows

Five sequential stages, each closed by a gate. A **CDISC data-standards track** (SDTM → ADaM → Define-XML + reviewer guides) runs through Stages 3–5, and a **hybrid-AI governance track** (data-classification routing, model-freeze records, engine-of-record register) runs across all five.

Stage	Focus	Gate
S1 — Study Design & Protocol	Biostatistics shapes the design: estimands, endpoints, sample-size rationale, randomization, and the statistical sections of the protocol.	Protocol approved / finalized
S2 — Planning & Specifications	Everything is specified before data: the SAP, TLF shells, ADaM/SDTM specs, the randomization schedule, the data-review and QC plans.	SAP & specifications final
S3 — Data & CDISC	Raw data becomes standardized, conformant, traceable CDISC datasets (SDTM → ADaM) with reviewer guides and Define-XML, under full data integrity controls.	Database lock
S4 — Analysis & QC	TLFs and PK/PD analyses are produced and independently double-programmed; reported numbers come from validated engines (Phoenix WinNonlin for NCA), never an LLM.	All outputs QC-passed
S5 — Reporting & CSR	Results become the CSR — statistical methods and results text, in-text tables/figures, appendices, the submission package, and cross-document consistency.	CSR draft delivered

S1 — Study Design & Protocol

Biostatistics shapes the design: estimands, endpoints, sample-size rationale, randomization, and the statistical sections of the protocol.

ID	Deliverable	Owner	Standard	QC	AI
S1-01	**Biostatistics input to protocol — design rationale & study design selection**	Lead biostatistician	ICH M11 (CeSHarP protocol template) design section; ICH E9 (design principles, bias	Independent biostatistician peer review; cross-check design vs objectives/estiman	GREEN

ID	Deliverable	Owner	Standard	QC	AI
			control); ICH E6(R3) GCP; FDA/EMA FIH guidance	ds; medical/clin-pharm/regulatory review; protocol team consensus	
S1-02	**Objectives → estimands → endpoints alignment (ICH E9(R1) estimand framework)**	Lead biostatistician	ICH E9(R1) estimand addendum; ICH E9; ICH M11 (estimand placement in protocol); ICH E3	Independent biostatistician review of estimand completeness (all 5 attributes, every ICE addressed); clinical & clin-pharm alignment review; consistency check objectives ↔ estimands ↔ endpoints ↔ analysis sets	GREEN
S1-03	**Statistical considerations / analysis-overview section of protocol**	Lead biostatistician	ICH E9 / E9(R1); ICH M11 statistical-considerations section; ICH E3 (consistency with future CSR methods); ICH E6(R3)	Peer review by independent biostatistician; alignment check vs estimands and vs planned SAP scope; cross-functional protocol review	GREEN
S1-04	**Analysis populations / analysis-set definitions (protocol-level)**	Lead biostatistician	ICH E9 / E9(R1); ICH M11; CDISC ADaMIG v1.3 (population-flag conventions, e.g., SAFFL/PKFL); ICH E3	Independent review of set definitions for mutual consistency and estimand linkage; clin-pharm review of PK-evaluable exclusion rules; traceability check to ADSL flag plan	GREEN
S1-05	**Sample-size / power justification (or documented rationale for no formal sizing in FIH)**	Lead biostatistician	ICH E9 (sample size & assumptions); ICH M11; FDA BA/BE & food-effect guidance; EMA bioequivalence/early-phase guidance; ICH E9(R1) (sizing tied to estimand)	Independent re-derivation/double-check of sample-size or precision calculation (independent code/program run); review of assumptions vs source data; documentation of method and seed/software version	AMBER
S1-06	**Randomization design & specification (incl. sentinel dosing & active:placebo allocation)**	Lead biostatistician	ICH E9 (randomization & blinding); ICH E6(R3) GCP; ICH M11; 21 CFR Part 11 (controlled, audit-trailed list generation); sponsor randomization SOP	Independent biostatistician review of design spec; verification that block/ratio/sentinel logic enforces intended allocation; segregation-of-duties check (spec	GREEN

ID	Deliverable	Owner	Standard	QC	AI
				author ≠ schedule generator); blinding-integrity review	
S1-07	**DSMB/SRC/SEC charter — statistical decision rules & dose-escalation criteria**	Lead biostatistician	ICH E9 (data monitoring, decision rules); ICH E6(R3) GCP; FDA guidance on Data Monitoring Committees; FIH guidance (escalation/stopping); ICH M11	Independent review of decision-rule logic for unambiguity and safety conservatism; consistency check vs protocol dose levels and stopping rules; committee/regulatory review; (if model-based) operating-characteristics simulation review	AMBER
S1-08	**Dose-escalation operating-characteristics simulation (design-stage)**	Lead biostatistician	ICH E9 / E9(R1); FDA adaptive-design & FIH guidance; reproducibility per 21 CFR Part 11 (versioned, seeded code); sponsor simulation SOP	Independent re-run/replication of simulation with documented seed & environment; code review; scenario-grid completeness review; sanity-check of OC against design intuition	AMBER
S1-09	**Interim / adaptive-design elements specification (if any)**	Lead biostatistician	ICH E9 / E9(R1); FDA Adaptive Designs for Clinical Trials guidance; EMA reflection paper on adaptive designs; ICH M11; 21 CFR Part 11	Independent statistical review of adaptation rules & error-rate control; simulation of operating characteristics; firewall/independence review; documentation of 'N/A' decision where no adaptation	AMBER
S1-10	**PK precision / variability assumptions memo (CV%, BLQ/LLOQ & sampling-scheme adequacy)**	Lead biostatistician	ICH E9; FDA/EMA PK and BA/BE guidance; sponsor NCA/PK SOP; reproducible assumptions	Joint biostatistician–PK scientist review; cross-check sampling adequacy vs predicted t1/2; sensitivity of precision to CV% assumptions; documentation of assumption sources	GREEN
S1-11	**Biostatistician protocol review & sign-off**	Lead biostatistician	ICH E9 / E9(R1); ICH M11; ICH E6(R3) GCP; 21 CFR Part 11 (electronic signature & audit	Comment-tracking & resolution log; independent (second-biostatistician)	GREEN

ID	Deliverable	Owner	Standard	QC	AI
			trail); sponsor protocol-approval SOP	consistency check on objectives ↔ estimands ↔ endpoints; signed/audited approval; amendment version control	
S1-12	**Emergency unblinding / code-break plan (design-stage definition)**	Lead biostatistician	ICH E6(R3) GCP; ICH E9 (blinding integrity); 21 CFR Part 11 (audit trail of unblinding); pharmacovigilance regs; sponsor unblinding SOP	Independent review of firewall & segregation-of-duties; verification unblinding pathway is auditable and minimally disruptive to the blind; cross-check vs DSMB charter and randomization spec	GREEN
S1-13	**Blinded Sample-Size Re-Estimation (BSSR) Plan — design-stage provision**	Lead biostatistician	ICH E9 / E9(R1); ICH E6(R3) GCP	Biostatistician review and sign-off; no unblinded data touched	GREEN
S1-14	**Estimand-to-design traceability & assumptions log (intercurrent-event strategy register)**	Lead biostatistician	ICH E9(R1) estimand framework	Biostatistician + medical reviewer sign-off	GREEN

Stage gate: Protocol approved / finalized. 14 deliverables.

S2 — Planning & Specifications

Everything is specified before data: the SAP, TLF shells, ADaM/SDTM specs, the randomization schedule, the data-review and QC plans.

ID	Deliverable	Owner	Standard	QC	AI
S2-01	**Statistical Analysis Plan (SAP)**	Lead biostatistician	ICH E9 + E9(R1) estimands, ICH E3 (alignment with CSR), ICH E6(R3) GCP, ICH M11; version-controlled, dated, signed	Senior biostatistician peer review; cross-check vs protocol & estimands; documented sign-off; formal version control of amendments	GREEN
S2-02	**SAP Amendment & Version-Control Log**	Lead biostatistician	ICH E9, ICH E3, ICH E6(R3), 21 CFR Part 11 (audit trail), ALCOA++	Independent review of each change; timestamp vs lock/unblinding verified; Part 11 audit trail retained	GREEN
S2-03	**Analysis-Set / Population**	Lead biostatistician	ADaMIG v1.3 (population flags),	Independent reviewer confirms	GREEN

ID	Deliverable	Owner	Standard	QC	AI
	Definitions		ICH E9(R1), CDISC CT	flag logic vs SAP; double-programmed in ADSL; reconciled at blinded data review	
S2-04	**TLF Shells / Mock-Up Shells (Tables, Listings, Figures)**	Lead biostatistician	ICH E3 (in-text/appendix structure), sponsor output standards, CDISC CT for terminology	Peer/medical-writer/clinical review of shells; titles/footnotes vs SAP; dummy-data render test for layout	GREEN
S2-05	**TLF Inventory / Table of Contents**	Lead biostatistician	ICH E3, sponsor standards	Cross-check vs SAP analyses (no orphan/missing outputs); peer review	GREEN
S2-06	**ADaM Dataset Specifications (ADSL)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1, traceability to SDTM	Spec peer review; downstream independent double-programming of ADSL; Pinnacle 21 ADaM conformance	GREEN
S2-07	**ADaM Dataset Specification (ADPC — PK Concentrations)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1, ICH PK reporting practice	Spec review vs PK plan (BLQ/time rules); independent double-programming of ADPC; reconciliation to SDTM PC	GREEN
S2-08	**ADaM Dataset Specification (ADPP — PK Parameters)**	Stat programmer	ADaMIG v1.3, CDISC CT (PK parameters), Define-XML v2.1	Spec review vs parameter list; ADPP reconciled to validated Phoenix output; independent double-programming of structure	AMBER
S2-09	**ADaM Dataset Specification (ADAE — Adverse Events)**	Stat programmer	ADaMIG v1.3 (OCCDS), MedDRA, CDISC CT, Define-XML v2.1	Spec review of TEAE logic; independent double-programming; reconcile vs SDTM AE	GREEN
S2-10	**ADaM Dataset Specifications (ADLB / ADVS / ADEG — Labs, Vitals, ECG)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1	Spec review of baseline/window/unit logic; independent double-programming; SDTM reconciliation	GREEN
S2-11	**ADaM Dataset**	Stat programmer	ADaMIG v1.3,	Spec review vs PD	GREEN

ID	Deliverable	Owner	Standard	QC	AI
	Specification (ADPD — Pharmacodynamics)		CDISC CT, Define-XML v2.1	plan; independent double-programming; SDTM reconciliation	
S2-12	**SDTM Mapping Specification**	Stat programmer	CDISC SDTMIG v3.4, Controlled Terminology, Define-XML v2.1	Independent review vs SDTMIG/CT; Pinnacle 21 rule mapping; traceability check raw->SDTM	AMBER
S2-13	**Annotated CRF (aCRF) — Preparation & Review**	Stat programmer	CDISC SDTMIG v3.4, Define-XML v2.1, CDISC aCRF conventions	Page-by-page review vs mapping spec; consistency with Define-XML origin; peer review	AMBER
S2-14	**Programming Specifications / Derivation Specs**	Stat programmer	ADaMIG v1.3, Define-XML v2.1 (value-level), CDISC CT	Independent spec review; basis for double-programming reconciliation; traceability to SAP	GREEN
S2-15	**PK Analysis Plan (NCA detail)**	PK scientist	ICH PK reporting practice, regulatory NCA guidance, Phoenix WinNonlin (validated NCA engine)	PK scientist + biostatistician review; rules cross-checked vs SAP; Phoenix template qualified	AMBER
S2-16	**popPK / Exposure-Response Modelling Analysis Plan (PMAP)**	Pharmacometrician	Regulatory popPK/E-R guidance, ICH E9(R1), software validation; NONMEM/Monolix as analysis engines	Independent pharmacometrics review; model-qualification criteria pre-specified; run records reproducible	GREEN
S2-17	**NONMEM/ Monolix Analysis Dataset Specification (PK/PD modelling dataset)**	Pharmacometrician	popPK dataset conventions, traceability to ADaM, software validation	Independent check vs ADaM source; structure/covariate completeness verified; reproducibility	AMBER
S2-18	**Randomization Schedule Generation**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11 (audit trail, access control), ALCOA++	Independent QC of allocation vs spec (ratios/blocks/strata); seed/reproducibility verified; secured storage	RED
S2-19	**Randomization Specification**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11	Independent review vs protocol; IRT UAT alignment; documented approval	GREEN
S2-20	**Unblinding / Code-Break Plan**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11 (audit trail),	SOP/charter alignment review; role/firewall	RED

ID	Deliverable	Owner	Standard	QC	AI
			ALCOA++	verification; emergency-unblinding log tested	
S2-21	**Data Review Plan / Blinded Data-Review Plan**	Lead biostatistician	ICH E9 (pre-unblinding decisions), ICH E6(R3), ALCOA++	Peer review; decisions recorded with timestamp before unblinding; consistency with SAP	GREEN
S2-22	**Define-XML Planning / Metadata Strategy**	Stat programmer	Define-XML v2.1, CDISC SDTMIG v3.4 / ADaMIG v1.3, Controlled Terminology	Metadata completeness vs specs; Pinnacle 21 define checks; value-level metadata review	AMBER
S2-23	**Controlled Terminology Plan / CT Version Pinning**	Stat programmer	CDISC Controlled Terminology, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1	Version-pin verification; Pinnacle 21 CT checks; extensible-codelist review	GREEN
S2-24	**Validation / QC Plan (Independent Double-Programming Strategy)**	Lead biostatistician	ICH E6(R3), 21 CFR Part 11, ALCOA++, CDISC conformance; sponsor validation SOPs	Risk-based assignment reviewed/approved; double-programming mandatory for key outputs; audit-ready	GREEN
S2-25	**Hybrid-AI Data-Classification & Routing Plan**	Lead biostatistician	21 CFR Part 11, ALCOA++, ICH E6(R3); sponsor AI/data-governance policy; HIPAA (API+BAA, not consumer)	Governance review of routing assignments; zero-egress verification for local tier; retention/audit-trail check	GREEN
S2-26	**Pseudo / Dummy Data for Shell Testing**	Stat programmer	CDISC structures (for realism), sponsor standards; non-PHI by construction	Structure conformance to ADaM specs; coverage of edge cases (BLQ, missing, windows)	AMBER
S2-27	**Data-Transfer Specifications (DTS / interface review)**	Data mgmt	CDISC SDTMIG v3.4 (target), 21 CFR Part 11, ALCOA++, sponsor DM SOPs	Test-transfer review; key/format/unit verification; reconciliation procedure agreed	GREEN
S2-28	**Dose-Escalation / Operating-Characteristics Simulation Plan & Outputs**	Lead biostatistician	ICH E9 + E9(R1), reproducibility/software validation	Independent review of assumptions/code; seed/reproducibility documented; results sanity-checked	GREEN
S2-29	**Interim Analysis Plan / DSMB statistical output specifications &**	Independent unblinded statistician + lead biostatistician	ICH E9; ICH E6(R3); FDA/EMA DSMB guidance	Independent unblinded statistician; firewall enforced	AMBER

ID	Deliverable	Owner	Standard	QC	AI
	firewall (SAP-level)				
S2-30	**Bioanalytical data-transfer & PK sample reconciliation specification (LC-MS/MS lab interface)**	PK scientist + biostat programming + bioanalytical lab	CDISC SDTMIG PC domain; FDA Bioanalytical Method Validation guidance; ALCOA++	Double-check vs lab dictionary; PK scientist sign-off	AMBER
S2-31	**Value-Level Metadata (VLM) specification for Define-XML**	Biostat programming / metadata lead	Define-XML v2.1; CDISC ADaMIG v1.3 / SDTMIG v3.4	Independent review vs datasets; Pinnacle 21 Define validation	AMBER
S2-32	**Software / computing environment validation & installation qualification (IQ/OQ) records**	Biometrics systems / QA + statistical programming	21 CFR Part 11; GAMP 5; ICH E6(R3)	QA review; reported numbers always from validated engines	GREEN

Stage gate: SAP & specifications final. 32 deliverables.

S3 — Data & CDISC

Raw data becomes standardized, conformant, traceable CDISC datasets (SDTM → ADaM) with reviewer guides and Define-XML, under full data integrity controls.

ID	Deliverable	Owner	Standard	QC	AI
S3-01	**EDC / raw data extracts & Data Transfer Agreement (DTA) interface**	Data mgmt (biostat/stat programmer co-reviewer)	21 CFR Part 11, ALCOA++, ICH E6(R3); CDISC CT for code lists; sponsor DTA SOP	Transfer-spec conformance check on each load; record counts & checksum reconciliation; test-transfer (dry-run) before first production transfer	GREEN
S3-02	**Biostat review of edit checks / data queries / data-review listings**	Lead biostatistician / Stat programmer (with Data mgmt)	ICH E6(R3) GCP, 21 CFR Part 11, ALCOA++; Data Review Plan	Independent review of listing logic; query resolution tracked to closure; reconciliation against SAP-defined critical variables	AMBER
S3-03	**Annotated CRF (aCRF)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4, Define-XML v2.1, Controlled Terminology; CDISC aCRF conventions	Peer review against final CRF and SDTM specs; Pinnacle 21 / Define linkage check that every annotated origin resolves	AMBER
S3-04	**SDTM datasets —**	Stat programmer	CDISC SDTMIG	Independent	RED

ID	Deliverable	Owner	Standard	QC	AI
	DM (Demographics)	(CDISC/SDTM lead)	v3.4, Controlled Terminology, Define-XML v2.1	double-programming + Pinnacle 21 conformance; reconciliation to raw and randomization	
S3-05	**SDTM datasets — PC (Pharmacokinetic Concentrations)**	Stat programmer (CDISC/SDTM lead) with PK scientist	CDISC SDTMIG v3.4 (Findings), Define-XML v2.1, value-level metadata; CT for units/analytes	Independent double-programming + Pinnacle 21; 100% reconciliation to bioanalytical transfer (record count, value, BLQ flags); nominal-vs-actual time check	RED
S3-06	**SDTM datasets — PP (Pharmacokinetic Parameters)**	Stat programmer (CDISC/SDTM lead) with PK scientist	CDISC SDTMIG v3.4 (PP domain), CT for PK parameters (PKPARMCD), Define-XML v2.1, RELREC linkage to PC	Reconciliation to Phoenix output (the engine of record); independent double-programming of the SDTM structure; Pinnacle 21	RED
S3-07	**SDTM datasets — EX (Exposure)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Interventions), Define-XML v2.1, Controlled Terminology	Independent double-programming + Pinnacle 21; reconciliation to randomization and to PC time-after-dose derivation	RED
S3-08	**SDTM datasets — AE (Adverse Events)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (MedDRA coding)	CDISC SDTMIG v3.4 (Events), MedDRA (versioned), Define-XML v2.1, CT	Independent double-programming + Pinnacle 21; MedDRA coding QC; reconciliation to safety database/SAE recon	RED
S3-09	**SDTM datasets — LB (Laboratory)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Findings), CT/LOINC, Define-XML v2.1 value-level metadata, unit standardization	Independent double-programming + Pinnacle 21; 100% reconciliation to lab transfer; unit conversion QC	RED
S3-10	**SDTM datasets — VS (Vital Signs)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to raw	RED
S3-11	**SDTM datasets — EG (ECG)**	Stat programmer (CDISC/SDTM lead) with PK/cardiac-	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1;	Independent double-programming +	RED

ID	Deliverable	Owner	Standard	QC	AI
		safety scientist	time-matched to PC for C-QTc	Pinnacle 21; reconciliation to ECG core-lab transfer; PK time-matching QC	
S3-12	**SDTM datasets — DS (Disposition)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Events), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to enrollment/randomization counts	RED
S3-13	**SDTM datasets — MH (Medical History)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (coding)	CDISC SDTMIG v3.4 (Events), MedDRA, CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC	RED
S3-14	**SDTM datasets — CM (Concomitant/Prior Medications)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (WHODrug coding)	CDISC SDTMIG v3.4 (Interventions), WHODrug (versioned), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC	RED
S3-15	**SDTM datasets — PD/DV and other supporting domains (PD/PE/SU/QS, DV, CO, SE/TA/TV/TE trial design, SUPPQUAL, RELREC)**	Stat programmer (CDISC/SDTM lead) with PD/PK scientists	CDISC SDTMIG v3.4 (Findings/Events/Special-Purpose/Trial-Design/SUPP/RELREC), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; RELREC integrity check (PC<->PP); deviation-classification reconciliation	RED
S3-16	**SDTM Define-XML (define.xml + stylesheet)**	Stat programmer (CDISC/SDTM lead)	CDISC Define-XML v2.1, SDTMIG v3.4, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; peer review; origin/aCRF cross-link check	AMBER
S3-17	**SDTM Reviewer Guide (cSDRG)**	Stat programmer (CDISC/SDTM lead) / Lead biostatistician	PHUSE cSDRG template, CDISC SDTMIG v3.4, Define-XML v2.1	Peer review; consistency with Define-XML and Pinnacle 21 explanations; sign-off	GREEN
S3-18	**ADaM datasets — ADSL (Participant-Level Analysis)**	Stat programmer (ADaM lead) / Lead biostatistician	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21 ADaM checks; population-flag reconciliation to disposition/randomization	RED
S3-19	**ADaM datasets — ADPC (PK Concentrations Analysis)**	Stat programmer (ADaM lead) with PK scientist	CDISC ADaMIG v1.3 (BDS / PK ADaM), Define-XML v2.1, traceability to PC/EX, CT	Independent double-programming + Pinnacle 21; time-after-dose & BLQ-rule QC;	RED

ID	Deliverable	Owner	Standard	QC	AI
				reconciliation to PC	
S3-20	**ADaM datasets — ADPP (PK Parameters Analysis)**	Stat programmer (ADaM lead) with PK scientist	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to PP/Phoenix, CT (PK params)	Reconciliation to Phoenix (engine of record); independent double-programming; Pinnacle 21	RED
S3-21	**ADaM datasets — ADAE (Adverse Events Analysis)**	Stat programmer (ADaM lead) / Lead biostatistician	CDISC ADaMIG v1.3 OCCDS, MedDRA, Define-XML v2.1, traceability to AE, CT	Independent double-programming + Pinnacle 21; TRTEMFL logic QC; reconciliation to SDTM AE & SAE database	RED
S3-22	**ADaM datasets — ADLB (Laboratory Analysis)**	Stat programmer (ADaM lead)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1 value-level metadata, traceability to LB, CT	Independent double-programming + Pinnacle 21; baseline/change & category-derivation QC; reconciliation to LB	RED
S3-23	**ADaM datasets — ADVS (Vital Signs Analysis)**	Stat programmer (ADaM lead)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to VS, CT	Independent double-programming + Pinnacle 21; baseline/change QC; reconciliation to VS	RED
S3-24	**ADaM datasets — ADEG (ECG Analysis)**	Stat programmer (ADaM lead) with cardiac-safety/PK scientist	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to EG/PC, CT; ICH E14/concentration-QTc practice	Independent double-programming + Pinnacle 21; QTc category & time-matching QC; reconciliation to EG	RED
S3-25	**ADaM datasets — ADPD and other analysis datasets (ADPD, ADCM, ADMH, deviation/exposure analysis as applicable)**	Stat programmer (ADaM lead) with PD/PK scientists	CDISC ADaMIG v1.3 BDS/OCCDS, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21; PK-PD linkage & baseline/change QC	RED
S3-26	**ADaM Define-XML (define.xml + stylesheet)**	Stat programmer (ADaM lead)	CDISC Define-XML v2.1, ADaMIG v1.3, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; derivation-method review; traceability check	AMBER
S3-27	**ADaM Reviewer Guide (ADRG)**	Stat programmer (ADaM lead) / Lead biostatistician	PHUSE ADRG template, CDISC ADaMIG v1.3, Define-XML v2.1	Peer review; consistency with Define-XML, SAP and Pinnacle 21 explanations; sign-	GREEN

ID	Deliverable	Owner	Standard	QC	AI
				off	
S3-28	**CDISC conformance / validation (Pinnacle 21) report — SDTM & ADaM**	Stat programmer (CDISC lead) / Lead biostatistician	CDISC conformance rules, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, FDA/PMDA validator rules	Validated Pinnacle 21 engine; documented triage/disposition of every finding; reviewer-guide reconciliation	AMBER
S3-29	**PK bioanalytical sample reconciliation (concentration data load & BLQ handling)**	PK scientist / Stat programmer (with bioanalytical lab & Data mgmt)	FDA/ICH M10 bioanalytical method validation, 21 CFR Part 11, ALCOA++; PK analysis plan BLQ rules	100% sample-accountability reconciliation; independent check of BLQ/LLOQ flagging; record-count/value match to transfer	AMBER
S3-30	**Blinded data review outputs (data-review listings & blinded review meeting package)**	Lead biostatistician (with Data mgmt, Clinical, PK)	ICH E9/E9(R1), ICH E6(R3); Data Review Plan; 21 CFR Part 11	Independent review of listings; documented blinded decisions (minutes) prior to unblinding; reconciliation to SAP rules	AMBER
S3-31	**Dry-run / test datasets & dry-run TLF execution**	Stat programmer / Lead biostatistician	Sponsor validation/QC plan; CDISC SDTMIG v3.4/ADaMIG v1.3; 21 CFR Part 11 (program version control)	Independent double-programming on dry-run; shell-vs-output reconciliation; defect tracking to closure	AMBER
S3-32	**Data cleaning / soft lock**	Data mgmt (with Lead biostatistician sign-off)	ICH E6(R3), 21 CFR Part 11, ALCOA++; sponsor lock SOP	Lock checklist completion; cross-functional reconciliation sign-off; audit-trail review	GREEN
S3-33	**Database hard lock**	Data mgmt (cross-functional sign-off incl. Lead biostatistician)	ICH E6(R3), 21 CFR Part 11 (audit trail/e-records), ALCOA++; sponsor lock SOP	Formal hard-lock checklist & multi-party sign-off; audit-trail/lock verification; snapshot integrity check	RED
S3-34	**Traceability (raw -> SDTM -> ADaM) matrix**	Stat programmer (CDISC lead) / Lead biostatistician	CDISC SDTMIG v3.4/ADaMIG v1.3/Define-XML v2.1 traceability principles; ICH E6(R3), ALCOA++	Cross-link verification raw<->SDTM<->ADaM; peer review; consistency with define/reviewer guides	AMBER
S3-35	**Controlled Terminology & coding-dictionary version management (CDISC CT, MedDRA, WHODrug)**	Data mgmt / Stat programmer (CDISC lead)	CDISC Controlled Terminology (versioned), MedDRA, WHODrug; Define-XML v2.1; 21 CFR Part 11	Version-lock verification; CT-conformance via Pinnacle 21; coding QC against locked dictionaries	GREEN

ID	Deliverable	Owner	Standard	QC	AI
S3-36	**Hybrid-AI data-classification routing log & engine-of-record register (Stage 3)**	Lead biostatistician (biometrics AI governance owner)	21 CFR Part 11, ALCOA++, ICH E6(R3); sponsor AI-governance SOP; hybrid-AI routing rules	Periodic governance review; routing-vs-classification reconciliation; model-freeze/version verification	GREEN
S3-37	**Randomization QC / verification report (schedule vs IXRS/EDC reconciliation)**	Independent statistician (not the schedule generator)	ICH E6(R3); 21 CFR Part 11; ALCOA++	Independent double-check; segregation of duties from generator	RED
S3-38	**Emergency unblinding / code-break event log & reconciliation**	Unblinded pharmacovigilance / IXRS admin + biostat	ICH E6(R3); 21 CFR Part 11 audit trail	Independent verification; segregation from blinded team	RED
S3-39	**Protocol-deviation classification & reconciliation log (important vs non-important)**	Clinical ops + biostat (blinded review)	ICH E3; ICH E6(R3); ICH E9(R1) (impact on populations/estimands)	Blinded review meeting; medical/biostat adjudication documented	AMBER
S3-40	**Analysis-population derivation & participant-disposition reconciliation (population-flag QC)**	Lead biostatistician + programming	CDISC ADaMIG v1.3 (ADSL population flags); ICH E9(R1)	Independent double-programming + blinded allocation sign-off	AMBER
S3-41	**Data Review Meeting minutes & blinded data-review sign-off (pre-lock)**	Cross-functional (DM, biostat, clinical, PK) — blinded	ICH E6(R3); ALCOA++; 21 CFR Part 11	Attendees confirm minutes; signatures retained	GREEN
S3-42	**Database lock authorization & 21 CFR Part 11 audit-trail review / system records**	Data management + QA + biostat	21 CFR Part 11; ALCOA++; ICH E6(R3)	QA review of audit trail; independent lock approval	GREEN
S3-43	**PK sampling-time deviation & actual-vs-nominal time reconciliation (PK timing dataset QC)**	PK scientist + biostat programming	CDISC PC/PP domains; FDA PK guidance; ALCOA++	Independent check vs source; PK scientist sign-off	AMBER

Stage gate: Database lock. 43 deliverables.

S4 — Analysis & QC

TLFs and PK/PD analyses are produced and independently double-programmed; reported numbers come from validated engines (Phoenix WinNonlin for NCA), never an LLM.

ID	Deliverable	Owner	Standard	QC	AI
S4-01	**Analysis environment qualification &**	Stat programmer	21 CFR Part 11; ALCOA++; ICH E6(R3)	Installation/ operational qualification	AMBER

ID	Deliverable	Owner	Standard	QC	AI
	program setup (folder structure, validated software versions, run/seed control)		(computerized systems / software validation); CSV/IQ-OQ-PQ	records; environment qualification checklist; version log; reviewer sign-off	
S4-02	**Disposition tables (screening, randomization, dosed, completed/discontinued; analysis-set attribution)**	Stat programmer	ICH E3; ADaMIG v1.3 (ADSL); SAP	Independent double-programming + reconciliation against shells; population counts cross-checked to ADSL	AMBER
S4-03	**Demographics & baseline characteristics tables**	Stat programmer	ICH E3; ADaMIG v1.3; CDISC CT; SAP	Independent double-programming + reconciliation; descriptive stats verified against ADSL/raw	AMBER
S4-04	**Exposure / dosing & treatment-compliance tables**	Stat programmer	ICH E3; SDTMIG v3.4 (EX); ADaMIG v1.3; SAP	Independent double-programming + reconciliation; dosing records cross-checked to EX/EC	AMBER
S4-05	**Protocol-deviation summary tables (with importance classification)**	Stat programmer	ICH E3; ICH E6(R3); SDTMIG v3.4 (DV); SAP	Independent double-programming + reconciliation; classification reviewed at blinded data review	AMBER
S4-06	**PK concentration summary tables + mean concentration-time figures (linear & semi-log)**	Stat programmer	ICH E3; ADaMIG v1.3 (ADPC); SDTMIG v3.4 (PC); SAP / PK analysis plan	Independent double-programming + reconciliation; BLQ handling verified; figures QC'd vs tables	AMBER
S4-07	**NCA execution in validated Phoenix WinNonlin (PK parameter derivation)**	PK scientist	Validated NCA in Phoenix WinNonlin; ICH E3; ADaMIG v1.3 (ADPP); SAP / PK analysis plan	Phoenix is the validated engine of record; independent re-derivation of key parameters (second analyst / second tool) + reconciliation; lambda_z selection reviewed; run records archived	RED
S4-08	**PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)**	Stat programmer	ICH E3; ADaMIG v1.3 (ADPP); SAP	Independent double-programming of the summary table + reconciliation;	AMBER

ID	Deliverable	Owner	Standard	QC	AI
				values traceable to validated Phoenix ADPP	
S4-09	**Dose-proportionality & accumulation analyses**	Stat programmer	ICH E3; SAP / PK analysis plan; relevant FDA/EMA dose-proportionality guidance	Independent double-programming + reconciliation; model specification reviewed; outputs traced to validated ADPP	AMBER
S4-10	**Food-effect / DDI / relative BA-BE statistical analysis (ANOVA, 90% CI, GMR) — if applicable**	Stat programmer	ICH E3; FDA/EMA BE & food-effect & DDI guidances; SAP	Independent double-programming of mixed-model/GMR /90% CI + reconciliation; model statement (fixed/random effects) reviewed; values traced to validated ADPP	AMBER
S4-11	**Concentration-QTc (C-QTc) / QT-QTc analysis — if applicable**	Pharmacometrician	ICH E14 + E14 Q&A (Q&A 5/6); ICH E3; ADaMIG v1.3 (ADEG); SAP	Independent re-modelling / double-programming + reconciliation; C-QTc model assumptions reviewed; run records archived	AMBER
S4-12	**popPK / exposure-response modelling (NONMEM/Monolix) with run records**	Pharmacometrician	Validated NONMEM/Monolix; popPK analysis plan; FDA/EMA popPK & exposure-response guidance; ICH E3	Run records (control streams/seeds/logs) archived; independent review of model code & assumptions; key results reproduced; VPC/GOF reviewed	RED
S4-13	**Safety TLFs — TEAEs (overview, by SOC/PT, severity, relationship, by dose)**	Stat programmer	ICH E3; MedDRA; ADaMIG v1.3 (ADAE); SAP	Independent double-programming + reconciliation; TEAE flag/derivation verified; counts cross-checked to ADAE	AMBER
S4-14	**Safety TLFs — deaths, SAEs, and AEs leading to discontinuation**	Stat programmer	ICH E3; ICH E2A/E6(R3); MedDRA; SAP	Independent double-programming + reconciliation; SAE/death flags reconciled with safety database/PV	AMBER
S4-15	**Safety TLFs —**	Stat programmer	ICH E3; CDISC CT;	Independent	AMBER

ID	Deliverable	Owner	Standard	QC	AI
	clinical laboratory analyses (summary, change-from-baseline, shift tables, outliers, Hy's law)		ADaMIG v1.3 (ADLB); SAP	double-programming + reconciliation; reference-range/grade derivations verified	
S4-16	**Safety TLFs — vital signs, ECG (12-lead parameters), physical exam**	Stat programmer	ICH E3; ICH E14 (ECG categorical); ADaMIG v1.3 (ADVS/ADEG); SAP	Independent double-programming + reconciliation; baseline/CFB and outlier criteria verified	AMBER
S4-17	**PD endpoint analyses (biomarker / target-engagement summaries and PK/PD)**	Pharmacometrician	ICH E3; ADaMIG v1.3 (ADPD); SAP	Independent double-programming + reconciliation; derivations/units verified; PK/PD models reviewed	AMBER
S4-18	**Immunogenicity / anti-drug antibody (ADA) analyses — if applicable**	Stat programmer	ICH E3; FDA/EMA immunogenicity guidance; CDISC IS; SAP	Independent double-programming + reconciliation; TE-ADA derivation verified	AMBER
S4-19	**Exploratory / post-hoc analyses**	Lead biostatistician	ICH E3; ICH E9; SAP (exploratory designation)	Risk-based QC (key exploratory outputs double-programmed); exploratory flagging reviewed	AMBER
S4-20	**Statistical analysis execution (run-all production of SAP-specified analyses)**	Stat programmer	ICH E9; ICH E3; 21 CFR Part 11; ALCOA++; SAP	Log review (no errors/warnings); output-vs-shell check; independent double-programming reconciliation gate before release	AMBER
S4-21	**Independent double-programming QC + reconciliation (discrepancy resolution)**	Stat programmer	ICH E6(R3); 21 CFR Part 11; ALCOA++; validation/QC plan; SAP	This IS the QC step — independent double-programming + documented reconciliation; lead biostatistician sign-off	AMBER
S4-22	**TLF validation / QC documentation & discrepancy log**	Stat programmer	21 CFR Part 11; ALCOA++; ICH E6(R3); validation/QC plan	Completeness/ consistency review of QC records; lead biostatistician sign-off	GREEN
S4-23	**Blinded / blind data-review meeting & documented decisions**	Lead biostatistician	ICH E9; ICH E6(R3); ICH E3; SAP / Data Review Plan	Decisions documented before unblinding; cross-functional sign-off; decisions	GREEN

ID	Deliverable	Owner	Standard	QC	AI
				locked	
S4-24	**Unblinding / code-break execution**	Lead biostatistician	ICH E6(R3); 21 CFR Part 11 (audit trail/access); unblinding/code-break SOP	Authorization & timing documented; access-controlled execution; audit trail; segregation of duties verified	RED
S4-25	**Pharmacometrics / popPK analysis report**	Pharmacometrician	FDA/EMA popPK & exposure-response guidance; ICH E3; modelling analysis plan	Independent review of model & report; key results reproduced; run-record completeness check	GREEN
S4-26	**QC documentation / validation records package (analysis stage)**	Lead biostatistician	21 CFR Part 11; ALCOA++; ICH E6(R3); validation/QC plan	Completeness/ traceability review; lead biostatistician sign-off; archived per retention policy	GREEN
S4-27	**Hybrid-AI governance log for analysis stage (data-classification routing, engine-of-record, model-freeze & prompt/output retention)**	Lead biostatistician	21 CFR Part 11 (audit trail/records); ALCOA++; internal hybrid-AI governance SOP	Routing/engine-of-record verified per output; freeze hashes recorded; retention completeness checked	GREEN
S4-28	**Phoenix WinNonlin NCA parameter-derivation QC / independent NCA verification record**	PK scientist + independent PK QC reviewer	Phoenix WinNonlin (validated NCA); FDA PK guidance; 21 CFR Part 11	Independent PK reviewer; documented parameter verification	RED
S4-29	**Pharmacometrics model qualification & validation records (covariate selection, GOF, VPC, bootstrap, run log)**	Pharmacometrician + independent reviewer	NONMEM/ Monolix; FDA popPK guidance; 21 CFR Part 11	Independent pharmacometrics review; reproducible run records	AMBER
S4-30	**Risk-based QC plan execution & QC-level assignment matrix (per-output QC tiering)**	Statistical programming QC lead	ICH E6(R3) risk-based quality management; ICH E8(R1)	QC lead sign-off; mandatory independent double-programming for RED outputs	AMBER
S4-31	**Concentration-QTc (C-QTc) model assumption-check & QC record**	Pharmacometrician / cardiac-safety statistician	ICH E14 / E14-S7B Q&A; FDA C-QTc guidance	Independent cardiac-safety statistician review	AMBER

Stage gate: All outputs QC-passed. 31 deliverables.

S5 — Reporting & CSR

Results become the CSR — statistical methods and results text, in-text tables/figures, appendices, the submission package, and cross-document consistency.

ID	Deliverable	Owner	Standard	QC	AI
S5-01	**CSR Statistical Methods section (Section 9.7 / 11) per ICH E3**	Lead biostatistician	ICH E3 (Section 9.7/11 structure), ICH E9 + E9(R1) estimands, CDISC ADaMIG v1.3 (population/derivation traceability)	Peer review by second biostatistician against final SAP; cross-check that every method maps to a reported output; medical-writer editorial QC; SAP-vs-CSR methods reconciliation checklist	GREEN
S5-02	**CSR Results — in-text (in-body) summary tables and figures**	Stat programmer	ICH E3 (Section 10–12 Results), CDISC ADaMIG v1.3 (source datasets), Phoenix WinNonlin (NCA engine of record)	Independent double-programming or 1:1 reconciliation of every in-text value against the corresponding full TLF and source ADaM; numerical-consistency check; QC log signed off	RED
S5-03	**Biostatistician contribution to and review of the CSR draft (Results text, consistency QC)**	Lead biostatistician	ICH E3, ICH E9/E9(R1), 21 CFR Part 11 (review records/audit trail)	Documented number-by-number consistency check (CSR-vs-TLF-vs-ADaM); second-reviewer spot-check; tracked-changes/comment resolution log retained as a controlled record	AMBER
S5-04	**CSR Appendix 16.1.9 — Documentation of statistical methods incl. final SAP**	Lead biostatistician	ICH E3 (Appendix 16.1.9), ICH E9/E9(R1), 21 CFR Part 11 / ALCOA++ (versioning, audit trail)	Completeness/version reconciliation (approved versions only); cross-check SAP content vs CSR Methods; QA review of appendix assembly	GREEN
S5-05	**CSR Appendix 16.2 — Data listings (by-participant)**	Stat programmer	ICH E3 (Section 16.2), CDISC SDTMIG v3.4 / ADaMIG v1.3	Independent double-programming or QC review of each listing; reconciliation against summary TLFs; QC log	RED
S5-06	**CSR appendices — Define-XML + reviewer guides**	Stat programmer	CDISC Define-XML v2.1, cSDRG/ADRG templates, CDISC	Pinnacle 21 conformance re-run on final	AMBER

ID	Deliverable	Owner	Standard	QC	AI
	(cSDRG/ADRG) and annotated CRF as submission appendices		Controlled Terminology, FDA/PMDA submission requirements	datasets; reviewer-guide completeness check; define-vs-dataset consistency QC	
S5-07	**Patient narrative data support (safety narratives)**	Stat programmer	ICH E3 (Section 12.3.2 / narratives), ICH E6(R3) GCP, 21 CFR Part 11, ALCOA++	Narrative-to-source data reconciliation per participant; independent check of the trigger list (all deaths/SAEs captured); QC log	AMBER
S5-08	**Participant (patient) profiles**	Stat programmer	ICH E3, CDISC SDTMIG v3.4 / ADaMIG v1.3, 21 CFR Part 11	Spot-check profiles vs source listings; independent review of profile program; reconciliation against ADaM	RED
S5-09	**Submission datasets + analysis programs as regulatory deliverables (with ADRG/cSDRG)**	Stat programmer	CDISC SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, Controlled Terminology; FDA Study Data Technical Conformance Guide; 21 CFR Part 11	Final Pinnacle 21 conformance; independent double-programming evidence retained; program executability/QC review; traceability raw→SDTM→ADaM→TLF confirmed	RED
S5-10	**Traceability matrix (raw → SDTM → ADaM → TLF/CSR)**	Lead biostatistician	CDISC ADaMIG v1.3 (traceability), Define-XML v2.1 value-level metadata, ICH E3, 21 CFR Part 11 / ALCOA++	Independent verification that each CSR/TLF number traces to a derivation and source; gap review; QA audit	AMBER
S5-11	**QC of CSR-vs-TLF-vs-dataset numerical consistency**	Lead biostatistician	ICH E3, ICH E9, 21 CFR Part 11 / ALCOA++ (QC records, audit trail)	Independent reviewer performs the check; discrepancy log tracked to closure; second-level QA sign-off	AMBER
S5-12	**Statistical input to clinical study synopsis and briefing documents**	Lead biostatistician	ICH E3 (synopsis), ICH E9/E9(R1), ICH M11 alignment for protocol/briefing context	Consistency check vs CSR body and TLFs (numbers identical); peer/medical review; sign-off	GREEN
S5-13	**Statistical/ biometrics sign-off and CSR release for submission**	Lead biostatistician	ICH E3, ICH E6(R3) GCP, 21 CFR Part 11 / ALCOA++ (e-signatures, audit trail), CDISC submission standards	Final QA review; confirmation all discrepancy logs closed; e-signature with audit trail; engine-of-record and model-freeze records attached	RED

ID	Deliverable	Owner	Standard	QC	AI
S5-14	**eCTD Module 5 submission package assembly & study-data standards conformance (study data reviewer's guide bundle, study-tagging)**	Regulatory + biometrics submission lead	FDA Study Data Technical Conformance Guide; eCTD; Define-XML v2.1	Submission-readiness review; Pinnacle 21 package validation	GREEN
S5-15	**Analysis Data Reviewer's Guide of record finalization & cross-document consistency (SAP ⇔ Define ⇔ ADRG ⇔ CSR)**	Lead biostatistician + metadata lead	ICH E3; ADaMIG v1.3; Define-XML v2.1; cSDRG/ADRG	Biostatistician sign-off; documented discrepancy closure	AMBER
S5-16	**Hybrid-AI governance dossier — model-freeze records, engine-of-record register, and prompt/output retention archive (submission-level)**	Biometrics AI-governance lead + QA	21 CFR Part 11; ALCOA++; ICH E6(R3); internal hybrid-AI policy	QA audit; segregation confirms LLM out of reported-number path	GREEN

Stage gate: CSR draft delivered. 16 deliverables.

CDISC dataset & metadata inventory

The standardized-data spine that makes the study submission-ready (SDTM → ADaM → Define-XML, with conformance and reviewer guides).

ID	Deliverable	Standard	QC
S3-04	SDTM datasets — DM (Demographics)	CDISC SDTMIG v3.4, Controlled Terminology, Define-XML v2.1	Independent double-programming + Pinnacle 21 conformance; reconciliation to raw and randomization
S3-05	SDTM datasets — PC (Pharmacokinetic Concentrations)	CDISC SDTMIG v3.4 (Findings), Define-XML v2.1, value-level metadata; CT for units/analytes	Independent double-programming + Pinnacle 21; 100% reconciliation to bioanalytical transfer (record count, value, BLQ flags); nominal-vs-actual time check
S3-06	SDTM datasets — PP (Pharmacokinetic Parameters)	CDISC SDTMIG v3.4 (PP domain), CT for PK parameters (PKPARMCD), Define-XML v2.1, RELREC linkage to PC	Reconciliation to Phoenix output (the engine of record); independent double-programming of the SDTM structure; Pinnacle 21
S3-07	SDTM datasets — EX (Exposure)	CDISC SDTMIG v3.4 (Interventions), Define-XML v2.1, Controlled Terminology	Independent double-programming + Pinnacle 21; reconciliation to randomization and to PC time-after-dose derivation
S3-08	SDTM datasets — AE (Adverse Events)	CDISC SDTMIG v3.4 (Events), MedDRA (versioned), Define-	Independent double-programming + Pinnacle 21;

ID	Deliverable	Standard	QC
		XML v2.1, CT	MedDRA coding QC; reconciliation to safety database/SAE recon
S3-09	SDTM datasets — LB (Laboratory)	CDISC SDTMIG v3.4 (Findings), CT/LOINC, Define-XML v2.1 value-level metadata, unit standardization	Independent double-programming + Pinnacle 21; 100% reconciliation to lab transfer; unit conversion QC
S3-10	SDTM datasets — VS (Vital Signs)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to raw
S3-11	SDTM datasets — EG (ECG)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1; time-matched to PC for C-QTc	Independent double-programming + Pinnacle 21; reconciliation to ECG core-lab transfer; PK time-matching QC
S3-12	SDTM datasets — DS (Disposition)	CDISC SDTMIG v3.4 (Events), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to enrollment/randomization counts
S3-13	SDTM datasets — MH (Medical History)	CDISC SDTMIG v3.4 (Events), MedDRA, CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-14	SDTM datasets — CM (Concomitant/Prior Medications)	CDISC SDTMIG v3.4 (Interventions), WHODrug (versioned), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-15	SDTM datasets — PD/DV and other supporting domains (PD/PE/SU/QS, DV, CO, SE/TA/TV/TE trial design, SUPPQUAL, RELREC)	CDISC SDTMIG v3.4 (Findings/Events/Special-Purpose/Trial-Design/SUPP/RELREC), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; RELREC integrity check (PC<->PP); deviation-classification reconciliation
S3-16	SDTM Define-XML (define.xml + stylesheet)	CDISC Define-XML v2.1, SDTMIG v3.4, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; peer review; origin/aCRF cross-link check
S3-17	SDTM Reviewer Guide (cSDRG)	PHUSE cSDRG template, CDISC SDTMIG v3.4, Define-XML v2.1	Peer review; consistency with Define-XML and Pinnacle 21 explanations; sign-off
S3-18	ADaM datasets — ADSL (Participant-Level Analysis)	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21 ADaM checks; population-flag reconciliation to disposition/randomization
S3-19	ADaM datasets — ADPC (PK Concentrations Analysis)	CDISC ADaMIG v1.3 (BDS / PK ADaM), Define-XML v2.1, traceability to PC/EX, CT	Independent double-programming + Pinnacle 21; time-after-dose & BLQ-rule QC; reconciliation to PC
S3-20	ADaM datasets — ADPP (PK Parameters Analysis)	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to PP/Phoenix, CT (PK params)	Reconciliation to Phoenix (engine of record); independent double-programming; Pinnacle 21
S3-21	ADaM datasets — ADAE (Adverse Events Analysis)	CDISC ADaMIG v1.3 OCCDS, MedDRA, Define-XML v2.1, traceability to AE, CT	Independent double-programming + Pinnacle 21; TRTEMFL logic QC; reconciliation to SDTM AE & SAE database

ID	Deliverable	Standard	QC
S3-22	ADaM datasets — ADLB (Laboratory Analysis)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1 value-level metadata, traceability to LB, CT	Independent double-programming + Pinnacle 21; baseline/change & category-derivation QC; reconciliation to LB
S3-23	ADaM datasets — ADVS (Vital Signs Analysis)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to VS, CT	Independent double-programming + Pinnacle 21; baseline/change QC; reconciliation to VS
S3-24	ADaM datasets — ADEG (ECG Analysis)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to EG/PC, CT; ICH E14/concentration-QTc practice	Independent double-programming + Pinnacle 21; QTc category & time-matching QC; reconciliation to EG
S3-25	ADaM datasets — ADPD and other analysis datasets (ADPD, ADCM, ADMH, deviation/exposure analysis as applicable)	CDISC ADaMIG v1.3 BDS/OCCDS, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21; PK-PD linkage & baseline/change QC
S3-26	ADaM Define-XML (define.xml + stylesheet)	CDISC Define-XML v2.1, ADaMIG v1.3, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; derivation-method review; traceability check
S3-27	ADaM Reviewer Guide (ADRG)	PHUSE ADRG template, CDISC ADaMIG v1.3, Define-XML v2.1	Peer review; consistency with Define-XML, SAP and Pinnacle 21 explanations; sign-off
S3-28	CDISC conformance / validation (Pinnacle 21) report — SDTM & ADaM	CDISC conformance rules, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, FDA/PMDA validator rules	Validated Pinnacle 21 engine; documented triage/disposition of every finding; reviewer-guide reconciliation
S3-31	Dry-run / test datasets & dry-run TLF execution	Sponsor validation/QC plan; CDISC SDTMIG v3.4/ADaMIG v1.3; 21 CFR Part 11 (program version control)	Independent double-programming on dry-run; shell-vs-output reconciliation; defect tracking to closure
S3-34	Traceability (raw -> SDTM -> ADaM) matrix	CDISC SDTMIG v3.4/ADaMIG v1.3/Define-XML v2.1 traceability principles; ICH E6(R3), ALCOA++	Cross-link verification raw<->SDTM<->ADaM; peer review; consistency with define/reviewer guides
S3-35	Controlled Terminology & coding-dictionary version management (CDISC CT, MedDRA, WHODrug)	CDISC Controlled Terminology (versioned), MedDRA, WHODrug; Define-XML v2.1; 21 CFR Part 11	Version-lock verification; CT-conformance via Pinnacle 21; coding QC against locked dictionaries
S3-43	PK sampling-time deviation & actual-vs-nominal time reconciliation (PK timing dataset QC)	CDISC PC/PP domains; FDA PK guidance; ALCOA++	Independent check vs source; PK scientist sign-off

TLF & analysis-output inventory (Stage 4)

Tables, listings, and figures by domain, plus the PK/PD analyses. Reported numbers are produced by validated engines, then independently double-programmed.

ID	Deliverable	Owner	QC
S4-02	Disposition tables (screening, randomization, dosed, completed/discontinued; analysis-set attribution)	Stat programmer	Independent double-programming + reconciliation against shells; population counts cross-checked to ADSL
S4-03	Demographics & baseline characteristics tables	Stat programmer	Independent double-programming + reconciliation; descriptive stats verified against ADSL/raw
S4-04	Exposure / dosing & treatment-compliance tables	Stat programmer	Independent double-programming + reconciliation; dosing records cross-checked to EX/EC
S4-05	Protocol-deviation summary tables (with importance classification)	Stat programmer	Independent double-programming + reconciliation; classification reviewed at blinded data review
S4-06	PK concentration summary tables + mean concentration-time figures (linear & semi-log)	Stat programmer	Independent double-programming + reconciliation; BLQ handling verified; figures QC'd vs tables
S4-07	NCA execution in validated Phoenix WinNonlin (PK parameter derivation)	PK scientist	Phoenix is the validated engine of record; independent re-derivation of key parameters (second analyst / second tool) + reconciliation; lambda_z selection reviewed; run records archived
S4-08	PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)	Stat programmer	Independent double-programming of the summary table + reconciliation; values traceable to validated Phoenix ADPP
S4-09	Dose-proportionality & accumulation analyses	Stat programmer	Independent double-programming + reconciliation; model specification reviewed; outputs traced to validated ADPP
S4-11	Concentration-QTc (C-QTc) / QT-QTc analysis — if applicable	Pharmacometrician	Independent re-modelling / double-programming + reconciliation; C-QTc model assumptions reviewed; run records archived
S4-12	popPK / exposure-response modelling (NONMEM/Monolix) with run records	Pharmacometrician	Run records (control streams/seeds/logs) archived; independent review of model code & assumptions; key results reproduced; VPC/GOF reviewed
S4-13	Safety TLFs — TEAEs (overview, by SOC/PT, severity, relationship, by dose)	Stat programmer	Independent double-programming + reconciliation; TEAE flag/derivation verified; counts cross-checked to ADAE
S4-14	Safety TLFs — deaths, SAEs, and AEs leading to discontinuation	Stat programmer	Independent double-programming + reconciliation; SAE/death flags reconciled with safety database/PV
S4-15	Safety TLFs — clinical	Stat programmer	Independent double-

ID	Deliverable	Owner	QC
	laboratory analyses (summary, change-from-baseline, shift tables, outliers, Hy's law)		programming + reconciliation; reference-range/grade derivations verified
S4-16	Safety TLFs — vital signs, ECG (12-lead parameters), physical exam	Stat programmer	Independent double-programming + reconciliation; baseline/CFB and outlier criteria verified
S4-18	Immunogenicity / anti-drug antibody (ADA) analyses — if applicable	Stat programmer	Independent double-programming + reconciliation; TE-ADA derivation verified
S4-22	TLF validation / QC documentation & discrepancy log	Stat programmer	Completeness/consistency review of QC records; lead biostatistician sign-off
S4-25	Pharmacometrics / popPK analysis report	Pharmacometrician	Independent review of model & report; key results reproduced; run-record completeness check
S4-28	Phoenix WinNonlin NCA parameter-derivation QC / independent NCA verification record	PK scientist + independent PK QC reviewer	Independent PK reviewer; documented parameter verification
S4-29	Pharmacometrics model qualification & validation records (covariate selection, GOF, VPC, bootstrap, run log)	Pharmacometrician + independent reviewer	Independent pharmacometrics review; reproducible run records
S4-31	Concentration-QTc (C-QTc) model assumption-check & QC record	Pharmacometrician / cardiac-safety statistician	Independent cardiac-safety statistician review

The hybrid-AI routing overlay

Every deliverable is risk-tiered (**Model Risk = influence × consequence**) and routed by data sensitivity. The pattern from the hybrid AI strategy applies here, deliverable by deliverable.

Tier	Count	What AI does	Engine
🟢 GREEN	48	Drafts, scaffolds, documentation, summaries on de-identified text	Cloud Claude (API + BAA) or local
🟡 AMBER	54	Dataset/TLF code; AI as **one side** of double-programming	Local frozen model (participant-level) / cloud (de-identified)
🔴 RED	34	**Nothing autonomous** — validated engine produces the number; LLM out of the path	Validated engine (Phoenix WinNonlin / SAS / NONMEM) only

The three hard rules (override everything): (1) PHI / unblinded / randomization data never leaves — always the local frozen model; (2) every **reported number** comes from a validated engine, never an LLM; (3) de-identification is the only thing that unlocks the cloud path (Claude API under BAA — never a consumer Copilot/Max seat).

Validation & governance (cross-cutting)

- **Independent double-programming** of every submission-grade output; the AI may replace at most one side, never both.
- **21 CFR Part 11 + ALCOA++**: secure, attributable, time-stamped audit trails; e-signatures; reproducibility.
- **Engine-of-record** recorded per deliverable; reported numbers always from a validated, qualified engine.
- **Model-freeze records** for the local open-weight model (fixed weights + config + seed = a re-runnable validation artifact); change control on any model update.
- **Data-classification routing log** and **zero-egress attestation** for sensitive data; consolidated into a submission-level hybrid-AI governance dossier.
- **Software / environment qualification** (IQ/OQ/PQ) for SAS/R, Phoenix WinNonlin, NONMEM/Monolix, Pinnacle 21.

Closing

This pipeline is the operating map for an in-house, hybrid-AI biometrics function: human-owned and signed at every step, validated engines for every reported number, full CDISC and Part 11 rigor — with AI accelerating the language, code-drafting, and QC-support layers under explicit, deliverable-by-deliverable controls. The complete register (every field, all 136 deliverables) is in the companion spreadsheet; the visual map is the accompanying one-pager.